

# Unit 6: CSS Flexbox

## Introduction to Flexbox

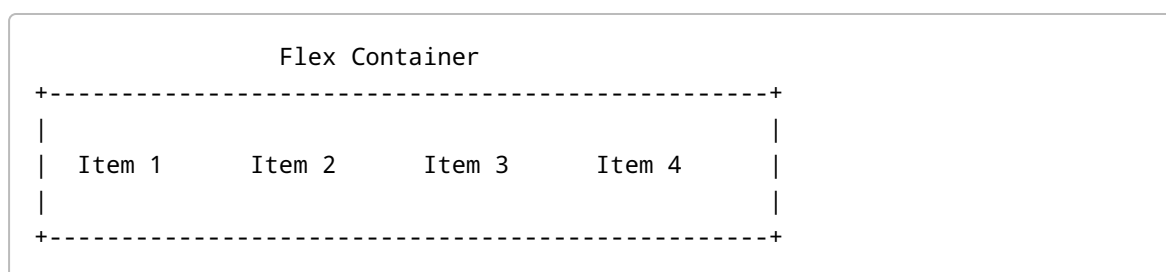
### Definition

**Flexbox (Flexible Box Layout)** is a CSS layout model used to arrange elements in rows or columns.

Flexbox works with two components:

- **Flex Container** – The parent element.
- **Flex Items** – The child elements inside the container.

### Diagram



## 1. display: flex

### Definition

The `display: flex` property converts a normal HTML element into a **Flex Container**. All child elements become **Flex Items** and are arranged in a row by default.

### Syntax

```
.container{  
  
display:flex;  
  
}
```

## Example

### HTML

```
<div class="container">

<div class="box">HTML</div>

<div class="box">CSS</div>

<div class="box">JavaScript</div>

</div>
```

### CSS

```
.container{

display: flex;

}

.box{

width: 100px;

height: 60px;

background: skyblue;

margin: 5px;

}
```

### Output

```
+-----+ +-----+ +-----+
| HTML | | CSS  | | JS   |
+-----+ +-----+ +-----+
```

---

## 2. flex-direction

### Definition

The `flex-direction` property specifies the direction in which flex items are displayed.

## Values

- row (Default)
- row-reverse
- column
- column-reverse

### Example 1 (Row)

```
.container{  
  
display:flex;  
  
flex-direction:row;  
  
}
```

## Output

HTML   CSS   JavaScript

---

### Example 2 (Column)

```
.container{  
  
display:flex;  
  
flex-direction:column;  
  
}
```

## Output

HTML  
CSS  
JavaScript

## Diagram

Row

```
+-----+ +-----+ +-----+  
|HTML| |CSS| | JS |  
+-----+ +-----+ +-----+
```

Column

```
+-----+
| HTML |
+-----+
| CSS  |
+-----+
| JS   |
+-----+
```

## 3. justify-content

### Definition

The `justify-content` property aligns flex items horizontally along the main axis.

### Common Values

- flex-start
- center
- flex-end
- space-between
- space-around
- space-evenly

### Example 1

```
.container{

display: flex;

justify-content: center;

}
```

### Output

HTML   CSS   JS

## Example 2

```
.container{  
  display:flex;  
  justify-content:space-between;  
}
```

## Output

|      |     |    |
|------|-----|----|
| HTML | CSS | JS |
|------|-----|----|

## Diagram

flex-start

HTML CSS JS

-----

center

HTML CSS JS

-----

space-between

HTML CSS JS

-----

space-around

HTML CSS JS

-----

space-evenly

HTML CSS JS

## 4. flex-wrap

### Definition

The `flex-wrap` property specifies whether flex items should move to the next line when there is not enough space inside the flex container.

By default, all flex items try to fit on a single line. Using `flex-wrap`, items can automatically wrap onto multiple lines, creating a more responsive layout.

### Values

- nowrap (Default)
- wrap
- wrap-reverse

### Example

```
.container{  
  
display:flex;  
  
flex-wrap:wrap;  
  
}
```

### Output

```
+-----+ +-----+ +-----+  
|HTML | | CSS | | JS |  
+-----+ +-----+ +-----+  
  
+-----+ +-----+  
|Java | |React|  
+-----+ +-----+
```

### Diagram

#### Without Wrap

```
HTML CSS JS Java React Python SQL C++  
-----> (Overflow)
```

#### With Wrap

HTML CSS JS

Java React Python

SQL C++

## 5. gap

### Definition

The `gap` property specifies the spacing between flex items.

### Syntax

```
.container{  
  gap:20px;  
}
```

### Example

```
.container{  
  display:flex;  
  gap:20px;  
}
```

### Output

```
+-----+   +-----+   +-----+  
| HTML |   | CSS  |   | JS  |  
+-----+   +-----+   +-----+
```

### Diagram

Without Gap

```
+-----+ +-----+ +-----+
|HTML| |CSS| |JS| |
+-----+ +-----+ +-----+
```

With Gap

```
+-----+   +-----+   +-----+
|HTML|   |CSS|   | JS|
+-----+   +-----+   +-----+
```